



CONTACT

LOCATION

Odesa, Ukraine · open to relocation

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GITHUB

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LINKEDIN

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TELEGRAM

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SKILLS

Frontend

TypeScript React Next.js Vue 3

Angular Redux Three.js

Tailwind

Backend

Node.js NestJS Express Hono

Django Flask Rails Spring Boot

ASP.NET Core REST GraphQL

gRPC

Real-time

WebRTC Socket.IO WebSocket

Yjs CRDT MQTT

Data & AI

Python NumPy Pandas

scikit-learn XGBoost TensorFlow

PyTorch Matplotlib

Databases

PostgreSQL MySQL MongoDB

SQLite Redis Firebase Drizzle

Security

Pentesting OWASP Top 10 WAF

Traffic analysis Burp Suite Nmap

Wireshark Snyc Semgrep

SonarQube

Tools & Games

Docker Git GitHub Actions Vite

Jest Unity / C#

Oleksii Silveistruk

Full-Stack Engineer · Applied ML & Cybersecurity Researcher

Full-stack developer who builds complete products end to end — from 3D graphics and real-time systems to ML-based security. Published researcher: a journal article on chess-engine search optimization and a conference paper on an ML Web Application Firewall (99.76% accuracy, 0.34% FPR). Master's student in cybersecurity with 74 public repositories on GitHub.

74

GITHUB REPOS

2

2026
PUBLICATIONS

120×

FEWER NODES
SEARCHED

99.76%

WAF ACCURACY

RESEARCH & PUBLICATIONS

JOURNAL ARTICLE · 2026

Optimization of the Minimax Algorithm with Alpha-Beta Pruning for Chess Artificial Intelligence Systems

O. R. Silveistruk · *Informatics. Culture. Technology*, Vol. 3, No. 1 (3), pp. 179–188 · Sep 2026 · in Ukrainian · [DOI 10.15276/ict.03.2026.15](#) · [Article page](#) · [PDF](#)

Python minimax engine extended with iterative deepening, Zobrist-hashed transposition tables, MVV-LVA move ordering, killer/history heuristics and quiescence search.

Evaluated nodes cut ~120× at depth 5; decision time 92.4 s → 0.43 s; up to 96.5% wins at level 0 against a reference engine.

CONFERENCE PAPER · AUG 2026

A System for Filtering HTTP Traffic of Web Applications (WAF) Using Machine Learning

O. R. Silveistruk · *Perspective Directions of Information Protection*, VIII Int. Sci.-Pract. Conf., Odesa, 26–28 Aug 2026, pp. 106–109 · in Ukrainian · [Proceedings \(PDF\)](#)

Intelligent reverse-proxy WAF: 51-feature request vectors (incl. Shannon entropy), SMOTE balancing, XGBoost / Random Forest / SVM / MLP ensemble with hybrid signature cross-validation. On 12,000 fuzzed transactions: **99.76% accuracy, 0.34% false positives, ~15 ms latency** (p99 14.8 ms at 1,500 RPS).

Earlier conference papers: [Current Trends in Scientific Research Development](#), Boston, Dec 2024 (pp. 269–274) · [European Congress of Scientific Achievements](#), Barcelona, Dec 2024 (pp. 206–210) · [79th SUITT Sci.-Tech. Conf.](#), Nov 2024 (pp. 185–187) · [80th SUITT Sci.-Tech. Conf.](#), Feb 2026 (pp. 296–300).

SELECTED PROJECTS

Online Compiler · own execution engine

2026

Browser IDE without Judge0/Piston: 41 languages, 615 reference snippets auto-checked against expected output, real-PTY interactive console, debugger for 7 languages, Scratch 3 / Blockly, Arduino Uno emulation in-browser and ESP32 in QEMU. CI + nightly runs.

[Django](#) [PostgreSQL](#) [Docker](#) [Monaco](#) [xterm.js](#) [WebSocket](#)

Chess 3D AI · basis of the journal article

2026

Four-package monorepo: custom TypeScript engine (alpha-beta, quiescence, opening book, 5 levels, Web Worker), procedural Three.js board with 3 themes and GSAP effects, Express + SQLite backend with Elo and analytics, Vue 3 client.

[TypeScript](#) [Three.js](#) [Vue 3](#) [Express](#) [SQLite](#)

CollabBoard · Miro-style whiteboard

2026

Real-time whiteboard + rich-text documents: 12 canvas tools, draw.io-style connectors with anchor snapping, Yjs CRDT collaboration, workspaces, admin panel, gateway over Socket.IO + gRPC.

[Next.js 16](#) [React 19](#) [Konva](#) [Slate](#) [Yjs](#) [gRPC](#) [Drizzle](#)

EDUCATION

M.Sc. Cybersecurity & Information Protection

State University of Intelligent Technologies and Communications
Sep 2025 – Dec 2026 (expected)

B.Sc. Computer Science

State University of Intelligent Technologies and Communications
Graduated 2025 · score 100
↗ Diploma

CERTIFICATIONS

EPAM

Front-End Fundamentals course certificate
↗ Verify certificate

Boston, Dec 2024

Participant, V Int. Sci.-Pract. Conf. “Current Trends in Scientific Research Development”
↗ Certificate ↗ Conference

Barcelona, Dec 2024

Participant, XII Int. Sci.-Pract. Conf. “European Congress of Scientific Achievements”
↗ Certificate ↗ Conference

ACTIVITIES

Science Skills Booster SUITT 2024

Key team member — “Ukrainian YouTube”
↗ Event report

Innovation Hub hackathons

Built and presented projects at the university
↗ Event report

Freelance

E-commerce websites for multiple clients

FOCUS AREAS

ML security

Game-tree search

Real-time systems

Developer tools

3D web

IoT

SELECTED PROJECTS (continued)

SocialNet

2026

Full social network in a pnpm monorepo: JWT + Google OAuth, feed with reactions, friends and people search, WebSocket messenger (voice, video notes, attachments), WebRTC audio/video calls, channels with roles, mini-games.

Next.js

Hono

Drizzle

WebSocket

WebRTC

Monitoring ESP32 · IoT + ML

2026

Sensor platform: MQTT ingestion from ESP32 devices, live dashboard, XGBoost anomaly detection and forecasting, seeded demo data and a one-command local stack.

Spring Boot 3.2

React 19

MQTT

XGBoost

MySQL

Peremoga · aid platform for Ukraine

2026

Donation and aid-coordination platform: four roles, tickets (offers/requests), fundraisers, news, live updates via ActionCable, ActiveAdmin, Swagger-documented API, UA/EN interface.

Rails 8.1

Angular 22

PostgreSQL 17

JWT

Music Platform · YouTube Music-style

2026

Streaming app with queue/shuffle/repeat player, playlists, artists and albums, genre/language search, cover tracks, JWT access + refresh, admin editing, Swagger.

NestJS 11

Next.js 16

MongoDB

Redux Toolkit

TanStack Query

LangLearn

2026

Language-learning platform: 18 courses in 5 languages, 8 lesson types, spaced-repetition review, progress tracking, admin CRUD, JWT with refresh tokens.

ASP.NET Core 9

EF Core

Angular 22

PostgreSQL

Mini Excel · no UI framework

2026

Spreadsheet in the browser on a hand-written mini-framework: formulas with reference shifting, charts, multiple sheets, merged cells, undo/redo, .xlsx import/export, 64 tests.

TypeScript

Vite

Soul Knight Clone & Lullaby of Echoes

2026

Unity 6 roguelike with procedural dungeons: 5 floors, 49 weapons, 60+ enemies, 6 bosses; all graphics and sound generated in code. Plus a narrative RPG with a “listen or erase” battle mechanic and 3 endings.

Unity 6

C#

Also on GitHub: [Video Chat \(WebRTC, NestJS\)](#) · [EduLink \(link shortener + LMS\)](#) · [КупиДевайс \(PERN e-shop\)](#) · [Shop \(Hono, Redis\)](#) · [North online bank \(Vue\)](#) · [Dela notes PWA](#) · [GitHub Explorer](#) · [Corporate Chats \(Flask\)](#) · [UVP project manager \(Django\)](#) · [Checkers AI \(8 rule sets\)](#).

EXPERIENCE

Junior Frontend Developer · Asferro

Jan – Feb 2022

Migrated an admin dashboard from GraphQL Apollo to REST; worked in a full-stack team until the company closed because of the war.

Production Intern · ISyb

Feb – Mar 2024

Built a full-stack auto-parts e-commerce site with React, Node.js and TypeScript.

Front-End Course · EPAM Systems ↗ certificate

Feb – Mar 2025

Completed the Front-End development course and earned a certificate.

“Ukrainian YouTube” · Science Skills Booster SUITT 2024

2024

Student content-sharing platform: React/Next.js (SSR/SSG), NestJS, PostgreSQL + TypeORM, JWT, video upload, categories, likes, comments, subscriptions.